

Table 1: **Task performance.** Task success, execution efficiency, and error detection across three environments. Results summarize three independent runs.

Assistant	CookSim				VHSim				ScreenSim			
	Success	Act.	Conv.	F1	Success	Act.	Conv.	F1	Success	Act.	Conv.	F1
<i>Frontier API models</i>												
GPT-6 Astra	26 $\pm$ 1	34	83	34	16 $\pm$ 4	14	90	58	90 $\pm$ 2	77	66	98
Gemini 3.1 Pro	21 $\pm$ 2	36	72	25	18 $\pm$ 3	13	86	75	86 $\pm$ 2	80	29	98
Gemini 3.8 Flash	31 $\pm$ 2	31	75	30	21 $\pm$ 6	15	80	68	79 $\pm$ 2	78	60	96
<i>Real-time streaming API models</i>												
Gemini 3.8 Live (polled)	22 $\pm$ 2	30	85	22	18 $\pm$ 3	25	80	22	72 $\pm$ 3	78	24	94
GPT-Realtime 2.1 (polled)	27 $\pm$ 2	40	1	22	19 $\pm$ 5	16	81	56	67 $\pm$ 2	78	5	98
Gemini 3.8 Live (native)	12 $\pm$ 5	48	90	1	—	—	—	—	—	—	—	—
GPT-Realtime 2.1 (native)	21 $\pm$ 8	43	45	7	—	—	—	—	—	—	—	—
<i>Open-weight non-streaming VLMs</i>												
Qwen3.8-27B	18 $\pm$ 1	37	35	17	25 $\pm$ 8	22	85	32	68 $\pm$ 2	78	13	96
Molmo2-8B	13 $\pm$ 1	39	2	2	3 $\pm$ 2	87	70	1	1 $\pm$ 1	87	0	16
InternVideo3-8B	14 $\pm$ 2	38	64	6	18 $\pm$ 4	34	81	0	0 $\pm$ 1	84	10	32
GLM-4.6V-Flash	12 $\pm$ 1	39	75	4	20 $\pm$ 2	26	82	21	0 $\pm$ 0	81	3	28
Kimi-VL-A3B-Thinking	15 $\pm$ 1	35	31	17	24 $\pm$ 4	25	82	65	10 $\pm$ 1	72	19	56
Muse Glimmer	24 $\pm$ 1	33	2	6	16 $\pm$ 2	17	73	20	44 $\pm$ 2	65	3	89
<i>Open-weight streaming VLMs</i>												
MiniCPM-o 4.5	14 $\pm$ 1	65	0	20	15 $\pm$ 2	26	59	14	3 $\pm$ 1	76	2	45
Qwen3-Omni-30B	26 $\pm$ 2	43	0	11	13 $\pm$ 2	46	65	3	15 $\pm$ 2	74	1	61
Proact-VL	11 $\pm$ 1	48	71	0	20 $\pm$ 3	35	63	0	0 $\pm$ 0	92	0	1
StreamingVLM	11 $\pm$ 2	45	78	0	19 $\pm$ 1	14	80	0	0 $\pm$ 0	91	88	1
<i>Reference conditions</i>												
No assistance	—	—	—	—	15 $\pm$ 5	11	100	0	0 $\pm$ 0	84	100	0
Scripted reference	—	—	—	—	—	—	—	—	90 $\pm$ 1	78	74	95

Scores are on a 0–100 scale. Success: in-time task completion (mean $\pm$ SD across three runs). Act. / Conv.: action / conversation efficiency; F1: error-detection F1 (three-run means). Bold: highest assistant success or F1 per environment. Dashes: unavailable results.